

BANSEOK KIM

☎ (437) 955-5109 ✉ bans.kim@mail.utoronto.ca 🏠 Toronto, Ontario

Education

University of Toronto

Bachelor of Applied Science in Civil Engineering + PEY Co-op

- Expected Graduation: 2028
- Recent Annual GPA: 3.87
- Awards: Dean's List - 2nd Year
- Relevant Courses: Steel and Timber Design, Reinforced Concrete, Structural Analysis, Solid Mechanics

Technical Skills

Software: SAP2000, VIC-3D, Revit, AutoCAD, Rhinoceros (Rhino), MS PowerPoint, MS Excel

Language: C, Python, Javascript

Availability: Available to work for 12 months or 16 months term

Work Experience

Structural Research Intern, University of Toronto

May 2025 - August 2025

Department of Civil and Mineral Engineering

- Constructed formwork, cut rebar, placed concrete, and cast specimens to support structural testing procedures
- Installed fibre optic sensors on steel specimens to enable experimental monitoring and real-time data collection
- Designed and executed an independent study to assess Digital Image Correlation (DIC) accuracy, producing quantifiable error metrics:
 - Achieved vertical/horizontal accuracy within ± 0.05 mm (2.2%) at 1 mm increments and 0.009 mm (9.6%) at 0.1 mm increments
 - Achieved rotational accuracy within $\pm 6\%$ error (absolute 0.2°)
 - Identified depth displacement sensitivity up to -28.9% at 0.1 mm increments
 - Improved precision by reducing horizontal error rates from 17.7% to 5.2% using shorter camera-to-object distances
- Delivered a technical report that advanced the lab's research goals and improved confidence in DIC technology

Relevant Engineering Projects

Project Manager, College Street Bikeways Extension

Jan 2024 - May 2024

- Led a 6-member team to develop a road design aligned with Toronto City Council's College Street Bikeway initiative
- Performed site assessments, stakeholder analysis, and traffic simulations to enhance cyclist safety and parking retention
- Ensured compliance with OTM Book 18 and Toronto's accessibility guidelines during all design phases
- Managed the full project lifecycle using Gantt charts, from ideation to stakeholder presentation

Wood Strength Testing Project

Mar 2025

- Conducted compressive and tensile strength testing on multiple wood species to analyze the effects of density, grain orientation, and moisture content
- Collected and interpreted experimental data to evaluate structural performance trends, including fiber-saturation point significance

EXTRA-CURRICULAR ACTIVITIES

Executive Member, University of Toronto Seismic Design Team (UTSD)

Sep 2024 - Present

- Modeled and optimized seismic-resistant tower structures in SAP2000, evaluating bracing systems and damper models to improve structural stability under earthquake loading
- Conducted shake table tests using various input motions (sine sweep, Kobe earthquake simulation, ground motion) to validate analytical models against experimental performance

Leadership Experience

Sergeant Squad Leader, Republic of Korea Army

Sep 2021 - Mar 2023

South Korea

- Supervised a squad of soldiers, ensuring operational readiness, discipline, and efficiency during daily military activities
- Coordinated logistics and personnel management, streamlining operations under strict timelines
- Served as an interpreter during joint exercises with United States Forces Korea (USFK), enabling clear communication between commanders